Reheating the KrF afterglow by inverse bremsstrahlung: the compendium in brief

The quantum derivation in short, the bottom line, and the efficiency that a kinetic model of the laser gas gives. Summary of *Inverse-bremsstrahlung heating of KrF laser gases* (the compendium, 163 pages), 20 October 2026

Bottom line.

- **Reheating is possible, and quantum mechanics says so.** The electrons of the afterglow absorb a THz wave, and at the rate of classical physics. The quantum rate of free–free absorption between exact scattering states reduces, for a photon that is soft against the electron's energy, to the momentum-transfer cross section, and the classical absorption built on it is right to **1.6 %** from 0.3 to 4 THz for electrons from 300 K to 1.08 eV (Sec. 2). Stimulated emission takes 0.2 % at the afterglow's 0.63 eV and 0.67 THz, the exchange of several photons changes no rate, and the nonlinearities of Langdon and Silin need 33 times the heating intensity and more. In the afterglow, 10^{20} m^{-3} at 0.63 eV, a 1.3-THz wave is absorbed through 10 cm of gas, and 1.4×10^{3} to 1.5×10^{4} W cm^{-2} holds the electrons at their temperature.
- **It does not make the laser meaningfully more efficient.** A time-dependent kinetic model of the Kr/F_2 laser gas, pumped with 300 kW cm^{-3} for 1 µs at 1 atm with 0.5 % F_2, which turns 5.73 % of the pump's energy into 248-nm light, was run with and without the compendium's 3-MW cm^{-2} wave in the afterglow (Sec. 3). With every rate at its default the wave *lowers* the efficiency, by 0.037 percentage points: each joule it gives the gas returns 4.5 % as 248-nm light, less than the pump's 5.73 %. It raises the efficiency only if Kr_2^* + F → KrF(B) + Kr, which no one has measured, runs faster than about 2×10^{-12} cm^3 s^{-1}, and then by at most 0.14 points, about 2 % of the efficiency, with a wave source at least 0.49 times as efficient as the pump's. Across that rate's band the change runs from **−0.04 to +0.14 points**, and the model's other uncertain inputs widen it to −2.57 to +0.35. Weaker waves lose 248-nm light, and with 2 % F_2 the afterglow's electrons last 8.5 ns, which leaves −0.0014 points.
- **What would decide it** (Sec. 4): a measurement of Kr_2^* + F → KrF(B) + Kr; the electrons' energy distribution under the THz field, on which the model's two closures disagree 24-fold in the absorbed energy; the stepwise ionization of Kr* and the electron quench of KrF*; and a model with a cavity, since the afterglow's 248-nm light is counted here as fluorescence.

1 The question

An electron beam or a field pumps a KrF laser gas for a microsecond. When the pump stops, the electrons it leaves behind cool and are lost to attachment and recombination: the afterglow. Reheating sends an electromagnetic wave into the afterglow, the wave heats the free electrons by inverse bremsstrahlung (IB), and the hotter electrons excite krypton again, which forms KrF* while F_2 lasts. The compendium asks whether that can make the laser more efficient. Its Part I splits the question into five links: the wave is absorbed through the gain medium (coupling), its absorption is known (absorption), it heats the electrons (heating), the heat forms KrF* (partition), and the KrF* becomes laser light that pays for the wave's source (laser and wall plug). The first three are physics that quantum mechanics settles, Section 2. The last two need the kinetics of the laser gas, Section 3.

2 Why the electrons absorb: the derivation in short

Gaussian units, as in the compendium. An electron of momentum $\hbar\mathbf{k}$ and energy $E = \hbar^2 k^2/2m$ scatters on an atom of potential $V(r)$ in a wave $E_0 \cos\omega t$ polarized along $\hat{\mathbf{e}}$, with $u = \hbar\omega/k_B T$ and N atoms and n_e electrons per unit volume.

(1) The golden rule. The rate of absorbing one photon between exact scattering states, $E \to E + \hbar\omega$, divided by the photon flux $cE_0^2/8\pi\hbar\omega$ and the electron flux, is the free–free cross section (Chapter 9, Eq. 9.14)

$$\sigma_+ = \frac{32\pi^2}{3}\,\frac{e^2 m^2 \omega}{\hbar^3 c}\,\frac{S}{k^3 k'}, \qquad S = \sum_l \left[(l+1)R^2_{l\to l+1} + l\,R^2_{l\to l-1}\right], \\ R_{l\to l'} = \frac{1}{m\omega^2}\int_0^\infty u_{k'l'}\,V'(r)\,u_{kl}\,\mathrm{d}r, \tag{1}$$

with u_{kl} the radial functions and $\hbar k' = (\hbar^2 k^2 + 2m\hbar\omega)^{1/2}$. The last form, the acceleration form, shows that the photon is absorbed where the atom accelerates the electron.

(2) Emission. Stimulated emission is the reverse transition, and detailed balance, $k^2\sigma_+(k \to k') = k'^2\sigma_-(k' \to k)$, leaves Maxwell electrons, with $g(E)$ their distribution of energies, the absorption coefficient (Eqs. 9.17 and 9.20)

$$\kappa = n_e N\,(1 - e^{-u})\int_0^\infty g(E)\,v\,\sigma_+(E)\,\mathrm{d}E. \tag{2}$$

(3) Soft photons. As $\hbar\omega/E \to 0$ the force identity $\int_0^\infty u_{k,l+1}\,V'\,u_{kl}\,\mathrm{d}r = (\hbar^2 k^2/2m)\sin(\delta_l - \delta_{l+1})$ (Eq. 9.31) turns S into the phase shifts, and σ_+ into the momentum-transfer cross section σ_m, the low-energy theorem of Low [1] (Eq. 9.34):

$$\sigma_+ \to \frac{4\pi}{3}\,\frac{e^2 v^2}{c\hbar\omega^3}\,\sigma_m(E), \qquad \sigma_m = \frac{4\pi}{k^2}\sum_l (l+1)\sin^2(\delta_l - \delta_{l+1}). \tag{3}$$

(4) The classical limit. With Eq. (3) and $1 - e^{-u} \to u$, Eq. (2) becomes the Drude absorption of a Lorentz gas (Eq. 9.37),

$$\kappa_{\mathrm{cl}} = \frac{\omega_p^2}{c\,\omega^2}\,\nu_{\mathrm{eff}}, \qquad \nu_{\mathrm{eff}} = \frac{mN}{3k_B T}\,\langle v^3 \sigma_m\rangle, \qquad \omega_p^2 = \frac{4\pi n_e e^2}{m}, \tag{4}$$

the energy-weighted collision frequency of the compendium's Eq. 4.12. The quantum and the classical absorption differ by the ratio (Eq. 9.39) $\mathcal{R} = \kappa/\kappa_{\mathrm{cl}} = [(1 - e^{-u})/u]\,\langle v^3\sigma_m G\rangle/\langle v^3\sigma_m\rangle$, with G the free–free Gaunt factor of the atom. Computed from the exact phase shifts of model potentials for neon, argon and krypton, $\mathcal{R}$ lies within 1.6 % of one from 0.3 to 4 THz at every electron temperature from 300 K to 1.08 eV in both fills (Sec. 9.9). Only far above that band does it matter: at 28.3 THz and 300 K the quantum absorption is 10 % below the classical one in the neon fill and 28 % below it in the argon fill.

(5) Ions. The same golden rule on the Coulomb potential gives the collision frequency $\nu_T\Lambda$ with $\Lambda = (\pi/\sqrt{3})\langle g\rangle(1 - e^{-u})/u$ (Eq. 4.15), which in the Born approximation is $\ln(4k_B T/e^\gamma\hbar\omega)$ for $u \ll 1$: the logarithm needs no cutoff put in by hand (Sec. 8.13). The ions matter only in a thermalized afterglow. At 300 K and $10^{20}\,\mathrm{m}^{-3}$ they absorb 1.4 times as much as the atoms in the neon fill at 1 THz, but while the electrons stay at 0.63 eV the ions catch up with the atoms only above $10^{22}\,\mathrm{m}^{-3}$ (Sec. 9.10).

(6) Strong fields. For electrons whose energy far exceeds the photon's, the cross section of Kroll and Watson [2] for exchanging ℓ photons heats at $\sum_\ell \ell^2 J_\ell^2(x) = x^2/2$, the one-photon rate, at any intensity (Sec. 9.7). The distribution's own nonlinearities need far more than heating does: Langdon's flattening [3] 33 to 51 times the heating intensity of the afterglow, and Silin's regime a further 11.5 times (Sec. 6.5).

(7) Heating. Each electron gains $P = e^2 E_{\mathrm{rms}}^2\,\nu_{\mathrm{eff}}/[m(\omega^2 + \nu_{\mathrm{eff}}^2)]$ (Eq. 4.7), and the field sets the electron temperature once P exceeds the electrons' collisional loss P_{loss}, at the intensity $I_{\mathrm{heat}} = P_{\mathrm{loss}}/\sigma_{\mathrm{abs}}$ with $\sigma_{\mathrm{abs}} = 4\pi e^2\nu_{\mathrm{eff}}/mc\omega^2$ for $\omega \gg \nu_{\mathrm{eff}}$ (Eq. 6.3).

In the afterglow. In the neon-buffered fill at 0.63 eV and $10^{20}\,\mathrm{m}^{-3}$ (Sec. 6.5, Table 6.2), a 0.67-THz photon has $u = 0.0044$, so stimulated emission takes $1 - (1 - e^{-u})/u = 0.2$ %, and the electrons' collision frequency is 0.12 of the wave's angular frequency. The wave is absorbed in 2.5 mm at 170 GHz and in 2.6 cm at 0.67 THz, so a depth of 10 cm takes 1.3 THz (Sec. 2.1). The electrons lose $3.6 \times 10^8\,\mathrm{eV\,s^{-1}}$

each, and I_{heat} is 1.4×10^3 W cm^{-2} at 170 GHz and 1.5×10^4 W cm^{-2} at 0.67 THz, within reach of gyrotrons.

So: quantum mechanics does not stand in the way. The afterglow's electrons absorb a THz wave at the classical rate, to 1.6 %, and the wave heats them at 10^3 to 10^4 W cm^{-2}.

3 How much it raises the efficiency

The formation chain overstates it. Counted with the branchings of a KrF kinetic network, a field far above the collision frequency forms 0.274 to 0.352 KrF* per absorbed joule at the afterglow's density, 1.4 to 1.8 times the electron beam's 0.198 (Secs. 2.5 and 6.8 of the compendium). That count assumes F_2 for every Kr* and leaves out the electron collisions of the excited atoms. A kinetic model that carries both answers differently.

The model. It is a time-dependent model of one uniform box of Kr/F_2 at constant volume, with 59 reactions on 12 species and an energy ledger that closes. The electrons' energy distribution is solved from the two-term Boltzmann equation with electron–electron collisions [4] and the wave's own high-frequency field term, so that the absorbed power is the solver's own. The atoms absorb through the classical kernel of Eq. (4) and the ions through the IB model of Turnbull et al. [5, 6]. Nothing in it is tuned to a device. Against the Tahoma amplifier it gives the net small-signal gain to -2 %, the saturation intensity to +4 % and the formation efficiency to -3 % (0.261 against 0.27), and its inputs' propagated uncertainty is ± 40 %.

The test. The gas, 1 atm of Kr with 0.5 % F_2 at 300 K, is pumped with 300 kW cm^{-3} put into its electrons for 1 µs and followed to 2 µs, once without and once with the compendium's wave in the afterglow: 47.6 kV cm^{-1} in amplitude, 3 MW cm^{-2}, at 6.2 THz (Sec. 4.9). Without the wave the gas turns 5.73 % of the pump's energy into 248-nm light. The *marginal* is the 248-nm energy that each absorbed joule of the wave adds, and the efficiency counted with the wave as an input rises only where the marginal beats the pump's 5.73 %. When the wave opens, the gas holds 2.7×10^{14} cm^{-3} electrons, at 0.60 eV on average over the afterglow.

gas; wave; window	absorbed, % of pump	marginal	pump's own	change, points	break-even source
1 atm, 0.5 % F_2; 3 MW cm^{-2}; 1–2 µs	3.1	4.5 %	5.73 %	-0.037	1.27
1 atm, 0.5 % F_2; 3 MW cm^{-2}; 1–1.1 µs	1.2	3.5 %	5.73 %	-0.026	1.62
1 atm, 0.5 % F_2; 0.3 MW cm^{-2}; 1–2 µs	0.055	0.81 %	5.73 %	-0.0027	7.1
1 atm, 0.5 % F_2; 0.03 MW cm^{-2}; 1–2 µs	0.004	-0.79 %	5.73 %	-0.0003	never
0.5 atm, 2 % F_2; 3 MW cm^{-2} at 6.6 THz; 1–2 µs	0.074	13.9 %	15.83 %	-0.0014	1.14

Table 1: The twin pairs with every rate at its default. The change is that of the 248-nm efficiency with the wave counted as an input. The break-even source is how efficient the wave's source, from electricity to absorbed power, must be relative to the pump's for the pair to break even at the wall plug: the pump's own efficiency over the marginal.

With 2 % F_2 at 0.5 atm the electrons last 8.5 ns after the pump, against 42 ns at 1 atm with 0.5 % F_2 (their density integrated over the afterglow, over its value when the wave opens), so the wave finds few electrons to heat.

Where the wave's energy goes. At 1 atm with the strong wave, 4.5 % of the absorbed energy leaves as 248-nm light, 29.5 % as the 147-nm light of Kr_2^*, 2.3 % in other bands, 52.8 % as heat, and 10.9 % stays in the gas as F atoms, Kr* and Kr_2^*. The electrons warm from 0.60 to 1.06 eV, and their time-integrated density grows 2.6-fold. But the pump has burned the F_2 below 10 % of its fill by 815 ns, before the wave opens, so the Kr* the wave makes finds no F_2 to harpoon. It forms Kr_2^*, which radiates at 147 nm, and more than half of the wave's energy ends as heat. The formation chain counts neither the burned F_2

nor these losses.

One unmeasured rate decides the sign. Kr_2^* + F → KrF(B) + Kr would hand that Kr_2^* to the F atoms the burn left. It is 0 in the model by default, the rule for a rate that no one has measured. Kannari et al. [7] use 3×10^{-10} cm^3 s^{-1}, an estimate carried over from Ar_2^* + F. Across its band:

Kr_2^* + F → KrF(B) + Kr, cm^3 s^{-1}	marginal	pump's own	change, points	break-even source	wave to 147 nm
0 (the default)	4.5 %	5.73 %	−0.037	1.27	29.5 %
1×10^{-12}	5.2 %	5.80 %	−0.018	1.11	28.0 %
3×10^{-12}	6.4 %	5.93 %	+0.014	0.92	25.4 %
1×10^{-11}	9.2 %	6.25 %	+0.077	0.68	18.9 %
3×10^{-11}	12.3 %	6.67 %	+0.127	0.54	10.5 %
1×10^{-10}	14.5 %	7.14 %	+0.138	0.49	4.0 %
3×10^{-10} (Kannari et al.)	15.3 %	7.45 %	+0.131	0.49	1.4 %
1×10^{-9} (the band's top)	15.6 %	7.67 %	+0.123	0.49	0.4 %

Table 2: The strong wave at 1 atm and 0.5 % F_2 over the whole afterglow, with Kr_2^* + F → KrF(B) + Kr across its band. The rate also raises the pump's own efficiency, since the pump's afterglow holds the same Kr_2^*.

The sign turns at about 2×10^{-12} cm^3 s^{-1}, under 1 % of Kannari's value. Above it the reheat gains at most 0.138 points of 7.14 %, about 2 % of the efficiency, and only with a wave source at least 0.49 times as efficient as the pump's.

The other inputs. Each of the model's 20 uncertain inputs at its two ends, one at a time, gives the change at the defaults as -0.037 ± 0.84 points in quadrature, and the envelope that stacks every input's excursions each way runs from −2.57 to +0.35 points. The measured 5s→5p cross section at the lower end of its band alone gives −1.55 points.

The other closure. With a Maxwellian closure of two temperatures in place of the solved distribution, the same wave sustains a discharge through the afterglow: the electrons reach 2.22 eV, the wave absorbs 75 % of the pump's energy, 85 % of it ends as heat, and the efficiency falls by 2.25 points.

Limits. The model is 0-D: the field inside the gas is given, and the gas 10 cm deep, the reflection and the absorption length as n_e falls are not modelled. It has no cavity. The 248-nm energy is fluorescence, and the afterglow's small-signal net gain rises with the wave from 0.12 to 0.20 % cm^{-1}, against 3.1 % cm^{-1} during the pulse, so there is little to extract after the pump either way. Waves stronger than 3 MW cm^{-2} were not run with the solved distribution. The pure-krypton model of the same family has not been run with the wave.

So: the reheat changes the 248-nm efficiency by −0.037 points at the model's defaults, by at most +0.14 points if the unmeasured Kr_2^* + F channel is fast, and by −0.0014 points with 2 % F_2. A gain of at most about 2 % of the efficiency, conditional on an unmeasured rate and on a THz source half as efficient as the pump, is not a reason to build the reheat.

4 What would settle it

1. **Kr_2^* + F → KrF(B) + Kr.** From about 1×10^{-11} cm^3 s^{-1} up the reheat pays at the model's reference point with a source 0.68 times as efficient as the pump's or better, and at 1×10^{-12} and below it does not pay at all (Table 2).
2. **The electrons' distribution under the THz field.** The model's two closures put the wave in different regimes and differ 24-fold in the energy it absorbs. A measurement, or a kinetic treatment that follows the distribution as the wave turns on, would settle it.
3. **The electron collisions of the excited atoms**: the stepwise ionization e + Kr* → Kr$^+$ + 2e and

the electron quench of KrF^*, which the compendium names as deciding the sign of a field's gain (Sec. 6.8). A THz pulse that sets the electron temperature and reads it back by its transmission can measure both.

4. **A cavity**, to count the extraction of the afterglow's KrF^*, and the wall plug with a real THz source at the frequency of Section 2.

References

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